

Digital Thinking & Innovation

Model Examination Paper

Time: 3 Hours

Full Marks: 60

Section A: Multiple Choice Questions (MCQs)

(Attempt all questions – 1×10 = 10 marks)

1. Which of the following best defines **digital thinking**?
2. a) Using computers only
3. b) Thinking creatively using digital tools and data
4. c) Writing programs
5. d) Using social media
6. Agile thinking mainly focuses on:
7. a) Fixed planning
8. b) Flexibility and continuous improvement
9. c) Hardware design
10. d) Risk avoidance
11. Which component is the brain of an expert system?
12. a) User Interface
13. b) Knowledge Base
14. c) Inference Engine
15. d) Database
16. Design thinking starts with:
17. a) Prototyping
18. b) Testing
19. c) Empathy
20. d) Implementation
21. Which AI area allows machines to understand human language?
22. a) Robotics
23. b) Computer Vision
24. c) NLP

25. d) IoT
26. Digital security mainly protects against:
27. a) Hardware failure
28. b) Cyber threats
29. c) Natural disasters
30. d) Power cuts
31. Which is NOT a digital business trend?
32. a) E-commerce
33. b) Digital payment
34. c) Manual bookkeeping
35. d) Remote work
36. Systems thinking focuses on:
37. a) Individual parts only
38. b) Whole system and relationships
39. c) Coding
40. d) Speed
41. Generative AI is used to:
42. a) Store data
43. b) Create new content
44. c) Encrypt data
45. d) Control networks
46. Culture change in the digital era is mainly driven by:
- a) Technology and globalization
 - b) Geography
 - c) Climate only
 - d) Population size

Section B: Short Answer Questions

(Attempt any FIVE – 5×4 = 20 marks)

1. Define digital thinking with one example.
2. Explain agile thinking in the digital era.

3. What is culture change? Mention two causes.
 4. Write the concept of an expert system.
 5. List and briefly explain any four categories of digital trends.
 6. What is digital security? Why is it important?
 7. Explain design thinking in short.
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Section C: Long Answer Questions

(Attempt any THREE – $3 \times 10 = 30$ marks)

1. Explain how the business world has been reshaped by digitalization with suitable examples.
 2. Describe the frontier of Artificial Intelligence. Explain at least five areas.
 3. Explain the relationship between agile thinking, digital thinking, design thinking, and systems thinking with a suitable example.
 4. Discuss culture change in the digital era. Include positive and negative impacts.
 5. Explain expert systems in detail. Describe its components and applications.
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Section D: Additional Multiple Choice Questions (MCQs)

(Practice Section)

1. Which thinking approach focuses mainly on rapid feedback and iteration?
 2. a) Systems thinking
 3. b) Design thinking
 4. c) Agile thinking
 5. d) Traditional thinking
6. Digital thinking mainly encourages:
 7. a) Manual work
 8. b) Technology-driven solutions
 9. c) Fixed procedures
 10. d) Paper-based systems
11. The main goal of design thinking is to:

- 12. a) Reduce cost only
- 13. b) Solve problems from a human perspective
- 14. c) Increase automation
- 15. d) Improve hardware

- 16. Systems thinking helps organizations to:
 - 17. a) Focus on individual tasks
 - 18. b) Ignore long-term impact
 - 19. c) Understand interconnections
 - 20. d) Work faster only

- 21. Which is an example of digital security?
 - 22. a) Social networking
 - 23. b) Two-factor authentication
 - 24. c) Online shopping
 - 25. d) Video streaming

- 26. Which of the following is a feature of agile thinking?
 - 27. a) Rigid planning
 - 28. b) Continuous improvement
 - 29. c) No customer feedback
 - 30. d) Long development cycles

- 31. Expert systems are mainly used to:
 - 32. a) Replace all humans
 - 33. b) Store large files
 - 34. c) Support decision making
 - 35. d) Design hardware

- 36. Which component stores rules and facts in an expert system?
 - 37. a) Inference engine
 - 38. b) User interface
 - 39. c) Knowledge base
 - 40. d) Compiler

- 41. Generative AI differs from traditional AI because it can:
 - 42. a) Only analyze data
 - 43. b) Only store information
 - 44. c) Create new content

45. d) Control networks
46. Which digital trend enables flexible working from home?
47. a) Cloud computing
48. b) Remote work platforms
49. c) Blockchain
50. d) Cyber laws
51. Culture change in organizations mainly affects:
52. a) Machines only
53. b) People, values, and behavior
54. c) Office buildings
55. d) Hardware systems
56. Digitalization in business mainly improves:
57. a) Manual paperwork
58. b) Speed, efficiency, and customer experience
59. c) Traditional advertising only
60. d) Physical storage
61. NLP is most closely related to:
62. a) Image processing
63. b) Human language understanding
64. c) Network security
65. d) Robotics movement
66. Which of the following best represents systems thinking?
67. a) Solving problems in isolation
68. b) Looking at cause-effect relationships
69. c) Ignoring feedback loops
70. d) Focusing on short-term goals only
71. Digital trends mainly influence:
72. a) Only IT companies
73. b) Only education sector
74. c) Society, business, and culture
75. d) Hardware manufacturing only
76. Ethical AI focuses on:

- 77. a) Faster algorithms only
 - 78. b) Transparency, fairness, and responsibility
 - 79. c) Entertainment
 - 80. d) Hardware efficiency

 - 81. An example of digital business model is:

 - 82. a) Street vending
 - 83. b) Subscription-based streaming service
 - 84. c) Manual accounting
 - 85. d) Paper marketing

 - 86. Which thinking approach ensures solutions are user-friendly?

 - 87. a) Agile thinking
 - 88. b) Systems thinking
 - 89. c) Digital thinking
 - 90. d) Design thinking
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Section E: Advanced & Scenario-Based MCQs

(Practice for Final Exam)

- 1. A company releases software in small updates after user feedback. This best represents:
- 2. a) Digital thinking
- 3. b) Systems thinking
- 4. c) Agile thinking
- 5. d) Traditional thinking

- 6. Using data analytics to predict customer behavior is an example of:

- 7. a) Design thinking
- 8. b) Digital thinking
- 9. c) Cultural thinking
- 10. d) Manual thinking

- 11. The first step in design thinking is to:

- 12. a) Test the solution
- 13. b) Define the problem
- 14. c) Empathize with users
- 15. d) Prototype quickly

- 16. Understanding how a change in one department affects the whole organization is:

- 17. a) Agile thinking
 - 18. b) Digital thinking
 - 19. c) Systems thinking
 - 20. d) Creative thinking
21. Which of the following best describes digital transformation?
- 22. a) Buying new computers
 - 23. b) Using email instead of letters
 - 24. c) Integrating digital technology into all business areas
 - 25. d) Creating social media accounts only
26. An expert system mainly depends on:
- 27. a) Hardware speed
 - 28. b) Human emotions
 - 29. c) Knowledge and rules
 - 30. d) Internet connection
31. Which is NOT a characteristic of an expert system?
- 32. a) Consistency
 - 33. b) Ability to explain decisions
 - 34. c) Human emotions
 - 35. d) Domain-specific knowledge
36. Facial recognition technology is an example of:
- 37. a) NLP
 - 38. b) IoT
 - 39. c) Computer Vision
 - 40. d) Blockchain
41. AI that learns by trial and error is called:
- 42. a) Supervised learning
 - 43. b) Unsupervised learning
 - 44. c) Reinforcement learning
 - 45. d) Deep storage
46. Protecting data using encryption relates to:
- 47. a) Digital marketing
 - 48. b) Digital security
 - 49. c) Digital culture

50. d) Digital creativity

51. Online collaboration tools mainly support:

52. a) Remote and hybrid work

53. b) Hardware repair

54. c) Manual reporting

55. d) Traditional offices

56. Which digital trend allows businesses to scale easily without owning infrastructure?

57. a) Blockchain

58. b) Cloud computing

59. c) Cyber law

60. d) Digital ethics

61. Culture change in the digital era is BEST seen in:

62. a) Slower communication

63. b) Increased paper usage

64. c) Flexible work culture

65. d) Fixed job roles

66. A system that suggests medical diagnosis based on symptoms is a:

67. a) Database system

68. b) Expert system

69. c) Transaction system

70. d) Operating system

71. Which thinking ensures long-term sustainability of solutions?

72. a) Agile thinking

73. b) Digital thinking

74. c) Systems thinking

75. d) Design thinking

76. Short-form videos and social media influence belong to:

77. a) Consumer digital trends

78. b) Security trends

79. c) Hardware trends

80. d) Network trends

81. Ethical AI is important mainly to:

- 82. a) Increase speed
 - 83. b) Reduce costs
 - 84. c) Ensure fairness and trust
 - 85. d) Improve graphics
86. Automation of repetitive office tasks shows:
- 87. a) Design thinking
 - 88. b) Cultural resistance
 - 89. c) Digital thinking
 - 90. d) Emotional intelligence
91. Which approach combines empathy, creativity, and testing?
- 92. a) Systems thinking
 - 93. b) Agile thinking
 - 94. c) Digital thinking
 - 95. d) Design thinking
96. Digital security helps organizations to:
- 97. a) Increase paper files
 - 98. b) Protect data and systems
 - 99. c) Slow down operations
 - 100. d) Avoid technology
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End of Question Paper